

Pembrolizumab in patients with advanced non-small-cell lung cancer (KEYNOTE-001): 3-year results from an open-label, phase 1 study



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Summary

Background The anti-programmed death 1 monoclonal antibody pembrolizumab has shown antitumour activity and is a first-line and second-line treatment option for patients with programmed death ligand 1 (PD-L1)-expressing advanced non-small-cell lung cancer. We report updated 3-year safety and efficacy outcomes from the phase 1 study, KEYNOTE-001.

Methods KEYNOTE-001 is a multicohort, open-label, phase 1 study of pembrolizumab (2 mg/kg every 3 weeks or 10 mg/kg every 2 or 3 weeks) in treatment naive or previously treated patients with locally advanced or metastatic non-small-cell lung cancer with measurable disease at baseline. Two cohorts were randomly assigned to a pembrolizumab dose by use of a computer-generated randomisation schedule at cohort-dependent ratios, and a further four cohorts were assigned to a pembrolizumab dose without randomisation. We present 3-year outcomes for the full analysis set of patients who received at least one dose of study treatment, pooled for all pembrolizumab doses. The primary efficacy endpoint was proportion of patients with objective response, analysed here as investigator-assessed response according to immune-related response criteria. Secondary efficacy endpoints included overall survival, duration of response, and progression-free survival. Safety endpoints included incidence of adverse events. This study is registered at ClinicalTrials.gov, number NCT01295827, and is ongoing.

Findings Between May 8, 2012 and July 13, 2014, 550 patients (101 treatment naive and 449 previously treated) were enrolled. Median follow-up was 34·5 months at data cutoff (Sept 1, 2016). At 36 months, investigator-assessed objective response according to immune-related response criteria was achieved for 41 of 101 treatment naive patients (41% [95% CI 30·9–50·8]; median duration of response was 16·7 months [95% CI 12·6–not reached]) and 102 of 449 previously treated patients (23% [18·9–26·9]; 33·3 [(22·5–not reached)]). The Kaplan-Meier estimate of overall survival at 36 months was 26·4% (95% CI 14·3–40·1) for treatment naive patients and 19·0% (15·0–23·4) for previously treated patients, with median overall survival of 22·3 months (95% CI 17·1–31·5) and 10·5 months (8·6–13·2). PD-L1 tumour proportion score $\geq 50\%$ was associated with longer median overall survival (95% CI) versus tumour proportion score 1–49% (treatment naive: 34·9 [20·3–not reached] vs 19·5 [10·7–26·3] months; previously treated: 15·4 [10·5–18·5] vs 8·5 [6·0–12·7] months). Grade 3–5 treatment-related adverse events occurred in 66 patients (12%), and 30 (6%) discontinued owing to a treatment-related adverse event. The most frequent grade 3–4 treatment-related adverse events were pneumonitis (10 [2%] of 550) and fatigue (5 [1%] of 550). Overall, 227 patients (41%) of 550 had serious adverse events, of which 50 (9%) were treatment related.

Interpretation Pembrolizumab provides durable response and long-term effects on overall survival, with tolerable safety, for treatment naive and previously treated patients with advanced non-small-cell lung cancer expressing PD-L1.

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Introduction

5-year survival for patients with non-small-cell lung cancer in the USA is low, especially for those with extra-thoracic metastases.¹ Despite availability of new treatment options, improvements in survival in non-small-cell lung cancer over the past 40 years have been relatively modest,¹ particularly for patients without actionable driver mutations.⁴ Although newer immunotherapies offer some promise with 30-month survival in previously treated patients of up to about 30%,⁵

long-term data, particularly in treatment naive patients, is scarce.

Cancers evade normal immune responses through multiple mechanisms, including using interactions between the programmed death 1 (PD-1) receptor and its ligands PD-L1 and PD-L2 to inhibit cytotoxic T cell responses. The humanised anti-PD-1 monoclonal antibody pembrolizumab, which blocks this interaction and thereby promotes tumour recognition and an enhanced antitumour immune response, has become an integral

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Research in context

Evidence before this study

Survival outcomes in patients with advanced non-small-cell lung cancer in the United States have improved moderately in the past 40 years, in part because of new chemotherapy regimens and targeted therapies. We used the National Comprehensive Cancer Network Clinical Practice Guidelines to review recommended treatment regimens and searched PubMed for clinical trials published in the last 10 years (from Aug 1, 2007, to Aug 1, 2017) on targeted therapies for NSCLC with terms that included “advanced NSCLC” and “pembrolizumab,” “EGFR,” or “ROS1”. At the time that the study was designed, there were no peer-reviewed publications describing outcomes with anti-programmed death (ligand) 1 (PD-[L]1) therapy in patients with advanced non-small-cell lung cancer.

Added value of this study

The ongoing KEYNOTE-001 study provides the longest available follow-up (median 34.5 months) with pembrolizumab treatment in advanced non-small-cell lung cancer and allows for comprehensive characterisation of overall survival, objective response, and safety.

Implications of all the available evidence

Since the initiation of KEYNOTE-001, from 2015 to 2018, several studies evaluating immunotherapy (specifically,

monotherapy) in patients with non-small-cell lung cancer have been published on pembrolizumab (eg, KEYNOTE-010, KEYNOTE-024, and KEYNOTE-042), nivolumab (eg, CheckMate 017 and CheckMate 057), and atezolizumab (eg, POPLAR). The pembrolizumab studies have shown significantly longer overall survival with pembrolizumab versus chemotherapy in patients with both treatment naive (KEYNOTE-042) and previously treated (KEYNOTE-010) non-small-cell lung cancer and PD-L1 tumour proportion score of 1% or greater, and patients with treatment naive non-small-cell lung cancer and PD-L1 tumour proportion score of 50% or greater (KEYNOTE-024). This updated analysis of the KEYNOTE-001 study provides additional long-term data, with a median follow-up of 34.5 months, showing that anti-programmed death 1 inhibition with pembrolizumab can lead to durable response and long-term overall survival in treatment naive and previously treated patients with advanced non-small-cell lung cancer expressing PD-L1. The studies on nivolumab and atezolizumab provide further support for durability of benefit with anti-PD-L1 therapy in patients with previously treated non-small-cell lung cancer with 3-year and 5-year follow-up; however, to date, long-term data are not available with these agents in treatment naive patients.

part of treatment for metastatic non-small-cell lung cancer. The antitumour activity and safety of pembrolizumab were first shown in locally advanced and metastatic non-small-cell lung cancer in the phase 1 KEYNOTE-001 study.^{6–8} After a median follow-up of 10.9 months (range 5.2–27.5), 25% of treatment naive patients had had an objective response (assessed by use of Response Evaluation Criteria in Solid Tumors version 1.1 [RECIST v1.1] per masked independent central review), as did 18% of previously treated patients. The study also validated PD-L1 expression as a biomarker associated with response during pembrolizumab therapy. The most frequently occurring treatment-related adverse events were fatigue, pruritus, and decreased appetite. There was no association between dose (pembrolizumab 2 mg/kg or 10 mg/kg every 3 weeks or 10 mg/kg every 2 weeks) and activity or toxicity.⁶

The efficacy and safety of pembrolizumab monotherapy were supported by findings of subsequent phase 3 randomised and open-label trials. In KEYNOTE-010,⁹ pembrolizumab showed significantly increased overall survival at both 2 mg/kg every 3 weeks (n=344; hazard ratio [HR] 0.71 [95% CI 0.58–0.88]; p=0.0008) and 10 mg/kg every 3 weeks (n=346; 0.61 [0.49–0.75]; p<0.0001) compared with docetaxel 75 mg/m² every 3 weeks (n=343) in patients with previously treated, PD-L1-positive (tumour proportion score ≥1%), non-small-cell lung cancer. In patients with a PD-L1 tumour proportion score of 50% or greater, progression-free survival and overall survival were

significantly increased for pembrolizumab compared with docetaxel (median follow-up 13.1 months [IQR, 8.6–17.7]).⁹ In KEYNOTE-024,¹⁰ significant improvements in progression-free survival and overall survival were shown with first-line pembrolizumab 200 mg every 3 weeks (n=154) versus platinum-based chemotherapy (n=151) in patients with advanced non-small-cell lung cancer with tumour proportion score of 50% or greater and without sensitising *EGFR* or *ALK* alterations (median follow-up 11.2 months [range, 6.3–19.7]).¹⁰ KEYNOTE-042 showed significantly improved overall survival with first-line pembrolizumab versus platinum-based chemotherapy in patients (n=637 each) with locally advanced or metastatic non-small-cell lung cancer without sensitising *EGFR* or *ALK* alterations and with tumour proportion score of 1% or greater (HR 0.81 [95% CI 0.71–0.93]).¹¹

We present 3-year efficacy and safety data from KEYNOTE-001 in patients with locally advanced (recurrent) or metastatic non-small-cell lung cancer treated with pembrolizumab. These data represent the most complete characterisation of efficacy and safety outcomes with pembrolizumab to date.

Methods

Study design and participants

KEYNOTE-001 is a multicohort, open-label, phase 1 study of pembrolizumab monotherapy for progressive, locally advanced or metastatic carcinoma (any type), melanoma, or non-small-cell lung cancer in treatment

naive and previously treated patients. The complete study design for the non-small-cell lung cancer cohorts has been previously reported.⁶

Study sites enrolled patients in the non-small-cell lung cancer cohorts if they were 18 years of age or older, had locally advanced or metastatic non-small-cell lung cancer, Eastern Cooperative Oncology Group performance status of 0 or 1, at least one measurable lesion per immune-related response criteria,¹² and a contemporaneous biopsy sample. For the treatment naive cohort, eligible patients had not received systemic therapy for advanced disease or adjuvant therapy within the previous year, had no *EGFR* or *ALK* alteration (except the first 11 patients enrolled on an earlier protocol version [protocol changed on March 28, 2013]), and had

tumours positive for PD-L1 expression. For the previously treated cohorts, eligible patients had received at least one or at least two previous systemic therapies for advanced disease, depending on the cohort (figure 1). Key exclusion criteria included autoimmune disease requiring systemic corticosteroids or immunosuppression and active central nervous system metastases (unless clinically stable for ≥ 4 weeks). The full list of inclusion and exclusion criteria are provided in the protocol (appendix).

The protocol and its amendments were approved by relevant institutional review boards or ethics committees (protocol number MK-3475-001-11). The study was done in accordance with the Good Clinical Practice guideline. All patients provided written informed consent.

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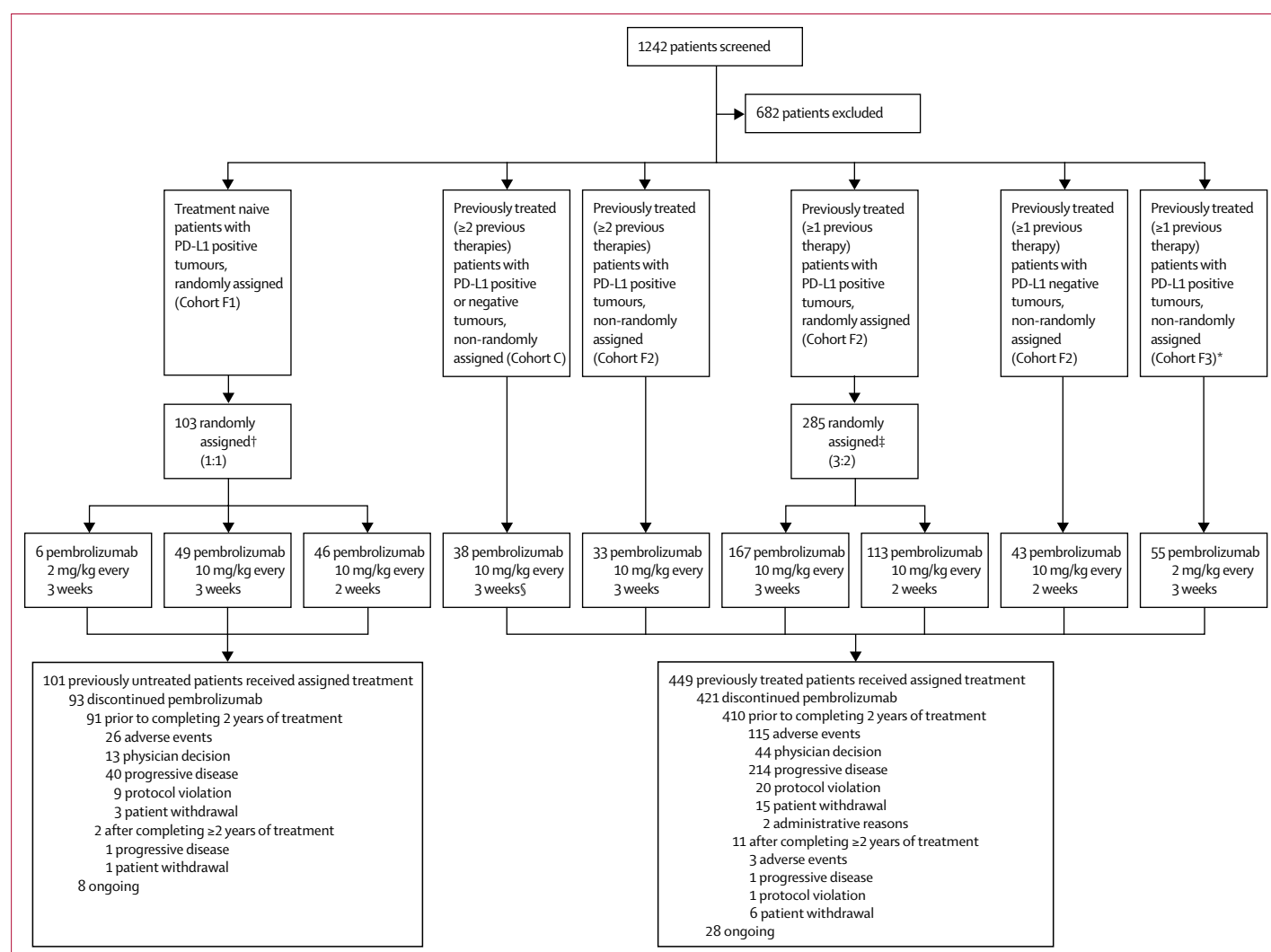


Figure 1: KEYNOTE-001 study profile

PD-L1=programmed death ligand 1. *Cohort F3 was added with protocol amendment 9 (Jan 31, 2014). †After randomisation in Cohort F1, 2 patients who were randomly assigned to 10 mg/kg every 3 weeks (n=1) and 10 mg/kg every 2 weeks (n=1) did not receive the assigned treatment due to an adverse event (n=1) and progressive disease (n=1); the first 11 patients enrolled in Cohort F1 were randomly assigned to pembrolizumab 2 mg/kg every 3 weeks or pembrolizumab 10 mg/kg every 3 weeks and the remaining 90 patients were randomly assigned to pembrolizumab 10 mg/kg every 3 weeks or pembrolizumab 10 mg/kg every 2 weeks. ‡After randomisation in Cohort F2, 5 patients randomly assigned to 10 mg/kg every 3 weeks did not receive the assigned treatment owing to adverse events (n=2), progressive disease (n=2), and screening failure (n=1). §41 patients were enrolled in Cohort C and 3 patients did not receive the assigned treatment owing to adverse events.

Randomisation and masking

Four cohorts were non-randomised (all previously treated patients) and two were randomised (one cohort of treatment-naïve patients and one of previously treated patients; see figure 1). In randomised cohorts, patients were assigned to treatment by the sponsor by use of a computer-generated simple randomisation schedule at cohort-dependent ratios. No minimisation or stratification was used for randomisation. The study was open-label; investigators and patients were not masked to the dose or schedule of pembrolizumab, and the allocation sequence was not concealed.

Procedures

Patients received intravenous pembrolizumab (Merck Sharp & Dohme, Whitehouse Station, NJ, USA) over 30 min at 2 mg/kg every 3 weeks, 10 mg/kg every 3 weeks, or 10 mg/kg every 2 weeks. Treatment continued until disease progression, intolerable toxicity, investigator decision, or patient withdrawal of consent. On the basis of protocol-specified guidelines (appendix), pembrolizumab was discontinued if an immune-mediated adverse event or infusion reaction did not resolve within 12 weeks of the last dose or if corticosteroids could not be reduced to less than or equal to 10 mg prednisone or equivalent per day within 12 weeks. Treatment was also discontinued following grade 3 or 4 or recurrent grade 2 pneumonitis; grade 4 diarrhoea or colitis; grade 3 or 4 aspartate aminotransferase elevation, alanine aminotransferase elevation, increased bilirubin, nephritis and renal dysfunction, or myocarditis; grade 3 or 4 hypophysitis or hyperthyroidism (at the investigator's discretion); grade 2 infusion reaction despite adequate premedication; anaphylaxis; and any other grade 4 or recurrent grade 3 immune-mediated adverse event or infusion reaction. Patients with confirmed complete response were allowed to discontinue treatment after 6 months or more and could resume treatment upon disease recurrence. A protocol amendment on April 20, 2016, allowed patients with partial response or stable disease to discontinue after at least 24 months; treatment could be resumed on disease progression or recurrence.

Imaging was done every 9 weeks. Response and disease progression were assessed by use of RECIST v1.1¹³ per masked independent central review (until April, 2016) and by use of immune-related response criteria per investigator review. Treatment decisions were based on immune-related response criteria¹² per investigator review. Survival was assessed every 2 months after treatment discontinuation. Adverse events during the study were recorded and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.0. PD-L1 status was assessed by immunohistochemistry by use of antibody clone 22C3 (Merck & Co, Kenilworth, NJ, USA). For enrolment, PD-L1 expression was assessed by use of a prototype assay (positivity defined as membranous staining on 1% or

more of tumour cells or distinctive staining pattern caused by adjacent mononuclear infiltrates).^{6,14} For analysis of efficacy outcomes by PD-L1 status, expression was assessed by use of a clinical trial assay (positivity defined as PD-L1 tumour proportion score $\geq 1\%$; stromal staining was not considered).¹⁵ Data were collected at study sites by use of the sponsor's electronic data capture and data management system.

Outcomes

The primary efficacy endpoint was the proportion of patients with objective response. To ease burden on sites and patients, central collection of imaging for independent central review of tumour response per RECIST v1.1¹³ was discontinued on April 20, 2016. Because this amendment occurred before our analysis, we report tumour response by immune-related response criteria¹² as evaluated by investigators. Before this amendment, concordance between independent central review and investigator review of tumour response was 92% (data not shown). Secondary efficacy endpoints included overall survival (time from treatment initiation to death due to any cause), duration of response (time from first confirmed response to disease progression), and progression-free survival (time from treatment initiation to disease progression or death due to any cause). This report provides an updated analysis of overall survival, objective responses by immune-related response criteria as evaluated by investigators, duration of response, progression-free survival, and safety.

Statistical analysis

Outcomes were assessed in all patients who received at least one study treatment dose. Because there was no association between pembrolizumab dose or schedule and efficacy or toxicity,⁷ dose groups were pooled for all analyses. Proportions of patients with objective response with 95% CIs were estimated by use of the binomial exact method. Overall survival and duration of response were estimated by use of the Kaplan-Meier method. Statistical analyses were done with SAS version 9.4 (SAS Institute, Cary, NC, USA). No data monitoring committee oversaw the study. This study is registered at ClinicalTrials.gov, number NCT01295827.

Role of the funding source

Representatives of the funder participated in study design, data collection, data analysis, data interpretation, and writing of the report. Data were collected by investigators and their site personnel. The sponsor maintained the study database. All authors had full access to all study data. Medical writing and editorial assistance were provided by C4 MedSolutions (Yardley, PA, USA), funded by Merck Sharp & Dohme Corp (Kenilworth, NJ, USA). All authors participated in the review and editing process, approved the submitted draft of the manuscript, had final responsibility for

	Treatment naive (n=101)	Previously treated (n=449)
Median (IQR) age (years)	68 (59–74)	62 (54–69)
Men	60 (59%)	229 (51%)
Women	41 (41%)	220 (49%)
ECOG performance status		
0	44 (44%)	148 (33%)
1	57 (56%)	299 (67%)
Unknown	0	2 (<1%)
Smoking history		
Current or former	90 (89%)	324 (72%)
Never	11 (11%)	125 (28%)
Histology		
Squamous	19 (19%)	76 (17%)
Non-squamous	79 (78%)	367 (82%)
Other or not specified	3 (3%)	6 (1%)
Molecular alterations		
EGFR mutation	n=98,* 3 (3%)†	n=429,* 74 (17%)
KRAS mutation	n=44,* 12 (27%)	n=283,* 74 (26%)
ALK translocation	n=94,* 1 (1%)†	n=392,* 8 (2%)
Previous systemic therapies‡		
0	93 (92%)	0
1	7 (7%)	81 (18%)
2	1 (1%)	139 (31%)
≥3	0	229 (51%)

Data are n (%) unless otherwise stated. ECOG=Eastern Cooperative Oncology Group. n=number of patients evaluated. *Excluding patients with unknown status. †Under the initial study protocol, testing for EGFR and ALK status was not required. A later protocol amendment on March 28, 2013, required EGFR and ALK testing and excluded treatment-naïve patients from enrolling if they had EGFR mutations or ALK translocation. ‡Includes (neo)adjuvant therapy.

Table 1: Patient demographics and baseline disease characteristics

	Treatment naive (n=101)	Previously treated (n=449)
Objective response	41 (41%)	102 (23%)
95% CI	30.9–50.8	18.9–26.9
Best overall response		
Complete response	2 (2%)	4 (1%)
Partial response	39 (39%)	98 (22%)
Stable disease	43 (43%)	160 (36%)
Progressive disease	10 (10%)	125 (28%)
No assessment	7 (7%)	62 (14%)
Median (IQR) time to response (months)	2.1 (2.0–4.1)	2.1 (2.0–4.1)
Median duration of response (months)	16.7	33.3
Range	2.1†–33.1†	1.0†–47.7†
95% CI	12.6–NR	22.5–NR
Patients with an ongoing response	19 (46%)	64 (63%)

Data are n (%) unless otherwise stated. NR=not reached. RECIST=Response Evaluation Criteria in Solid Tumors. *Response per immune-related response criteria per investigator assessment (rather than per RECIST v1.1 per independent central review) is reported for this analysis because central assessment of response ceased in 2016. †Ongoing response.

Table 2: Objective response by immune-related response criteria per investigator assessment*

the decision to submit for publication, vouch for the accuracy and completeness of the data reported, and attest that the study was done in accordance with the protocol.

Results

Between May 8, 2012, and July 13, 2014, 550 patients (101 treatment naïve, 449 previously treated) with non-small-cell lung cancer were enrolled and received treatment (figure 1). At data cutoff (Sept 1, 2016), median follow-up was 34.5 months (IQR 32.2–37.4). Median (IQR) duration of treatment at data cutoff for this analysis was 3.3 months (1.4–9.2) versus 3.3 months (1.4–9.2) on Sept 18, 2015 and 2.8 months (1.4–7.0) on Sept 1, 2014. At cutoff for this analysis, 36 patients (7%; eight treatment naïve, 28 previously treated) continued treatment; seven (one treatment naïve and six previously treated) received treatment beyond disease progression. 48 patients (9%; ten treatment naïve and 38 previously treated) completed 2 years or more of therapy; 35 (eight treatment naïve and 27 previously treated) chose to continue study treatment with pembrolizumab. Ten patients (one treatment naïve

and nine previously treated) completed 3 years or more of treatment. At data cutoff, 137 patients (37 [37%] of 101 treatment naïve and 100 [22%] of 449 previously treated) were alive. Patient demographics and baseline disease characteristics are summarised in table 1.

Objective response, as assessed by investigators according to immune-related response criteria, was achieved in 41 (40.6% [95% CI 30.9–50.8]) treatment naïve patients and 102 (22.7% [18.9–26.9]) previously treated patients; six patients had complete response (two treatment naïve, four previously treated). Median time to response was 2.1 months in both groups. Median duration of response was 16.7 months (95% CI 12.6–not reached) in treatment naïve patients and 33.3 months (22.5–not reached) in previously treated patients (table 2). The proportion of responding patients with ongoing response at data cutoff was 46% in treatment naïve patients and 63% in previously treated patients (table 2). Timing of objective response and progression are shown in figure 2. Most responses occurred within 3 months of treatment (33 [81%] of 41 treatment naïve; 71 [70%] of 102 previously treated), with nearly all responses occurring by 9 months (40 [98%] of 41 treatment naïve and 101 [99%] of 102 previously treated). Additionally, in the treatment naïve group, one patient had a partial response at 9.5 months and two had subsequent complete response at 10 months and 18 months. Among the previously treated patients, one patient had a partial response at 12.2 months, and five had subsequent complete response between 4.2 and 33.0 months (figure 2).

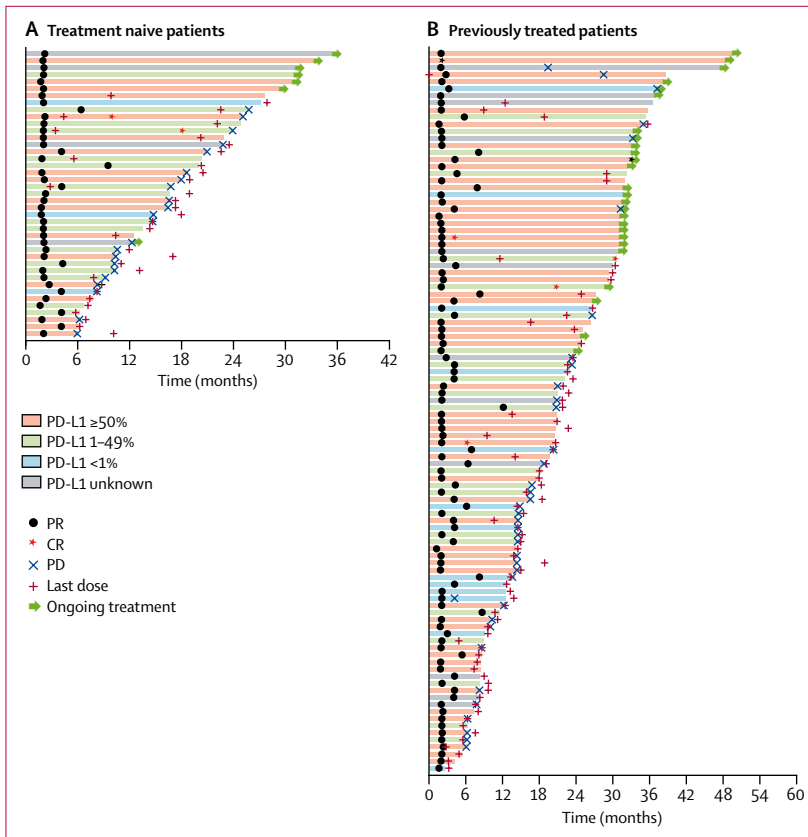


Figure 2: Time to response and time to progression by immune-related response criteria per investigator assessment*
(A) Treatment naïve patients. (B) Previously treated patients. CR=complete response. PD=progressive disease. PD-L1=programmed death ligand 1. PR=partial response. RECIST=Response Evaluation Criteria in Solid Tumors. *Response per immune-related response criteria per investigator assessment (rather than per RECIST v1.1 per independent central review) is reported for this analysis because central assessment of response ceased in 2016.

Objective response was achieved in 18 (66.7% [95% CI 46.0–83.5]) of 27 treatment naïve patients with PD-L1 tumour proportion score of 50% or greater and 16 (30.8% [18.7–45.1]) of 52 treatment naïve patients with tumour proportion score 1–49%. Objective response was achieved in 54 (39.1% [30.9–47.8%]) of 138 previously treated patients with tumour proportion score of 50% or greater, in 23 (13.7% [8.9–19.8%]) of 168 previously treated patients with tumour proportion score 1–49%, and in 13 (14.4% [7.9–23.4]) of 90 previously treated patients with tumour proportion score less than 1%. Only three treatment naïve patients had *EGFR* mutations, which precluded further response analyses. Objective response was achieved in four (5.4% [95% CI 1.5–13.3]) of 74 previously treated patients with *EGFR* mutations. Overall, 58 patients with CNS metastases were included in this analysis. In this population, objective responses were achieved in none of the three treatment naïve patients (95% CI 0.0–70.8) and in 12 of the 55 previously treated patients (21.8% [11.8–35.0]); 30 of the 58 patients had progressive disease (one treatment naïve, 29 previously treated), four of which

	Treatment-naïve (n=10)	Previously treated (n=38)
Best overall response		
Complete response	0	2 (5%)
Partial response	8 (80%)	32 (84%)
Stable disease	2 (20%)	4 (11%)
Progressive disease	0	0
No assessment	0	0
Median duration of response (months)	NR	NR
Range	10–33†	18–48†
95% CI	10.2–NR	31.2–NR

Data are n (%) unless otherwise stated. NR=not reached. RECIST=Response Evaluation Criteria in Solid Tumors. *Response per immune-related response criteria per investigator assessment (rather than per RECIST v1.1 per independent central review) is reported for this analysis because central assessment of response ceased in 2016. †Ongoing response.

Table 3: Confirmed best overall response by immune-related response criteria per investigator assessment* in patients who completed 24 months of treatment with pembrolizumab

were due to new lesions in the brain. Cerebral progression (new lesion) in patients who did not have CNS metastases at baseline occurred in four (4%) of 98 treatment naïve patients and 22 (6%) of 394 previously treated patients.

A majority of the 48 patients who completed 2 years or more of pembrolizumab therapy at data cutoff had a partial response as the best overall response (complete response, n=2; partial response, n=40; stable disease, n=6; table 3). Although the protocol amendment on April 20, 2016 allowed patients with partial response or stable disease to discontinue after 24 months or more, 35 of these patients chose to continue study treatment with pembrolizumab.

In our analysis, 64 (63%) of 101 treatment naïve and 349 (78%) of 449 previously treated patients had died. The Kaplan-Meier estimate of overall survival at 36 months was 26.4% (95% CI 14.3–40.1) and 19.0% (15.0–23.4), respectively. Median overall survival was 22.3 (17.1–31.5) and 10.5 (8.6–13.2) months (figure 3).

Among the 27 treatment naïve patients with PD-L1 tumour proportion score of 50% or greater, median overall survival was 34.9 months (95% CI 20.3–not reached); 24-month overall survival was 66.7% (95% CI 45.7–81.1), and 36-month survival was 25.2% (95% CI 5.0–53.1). For the 52 treatment naïve patients with tumour proportion score of 1–49%, median overall survival was 19.5 months (95% CI 10.7–26.3). There were too few treatment naïve patients with tumour proportion score less than 1% (n=12) for interpretation. Among the 138 previously treated patients with tumour proportion score of 50% or greater, median overall survival was 15.4 months (95% CI 10.5–18.5); 24-month overall survival was 38.6% (95% CI 30.4–46.7), and 36-month overall survival was 29.7% (95% CI 21.9–37.9). Median overall survival was

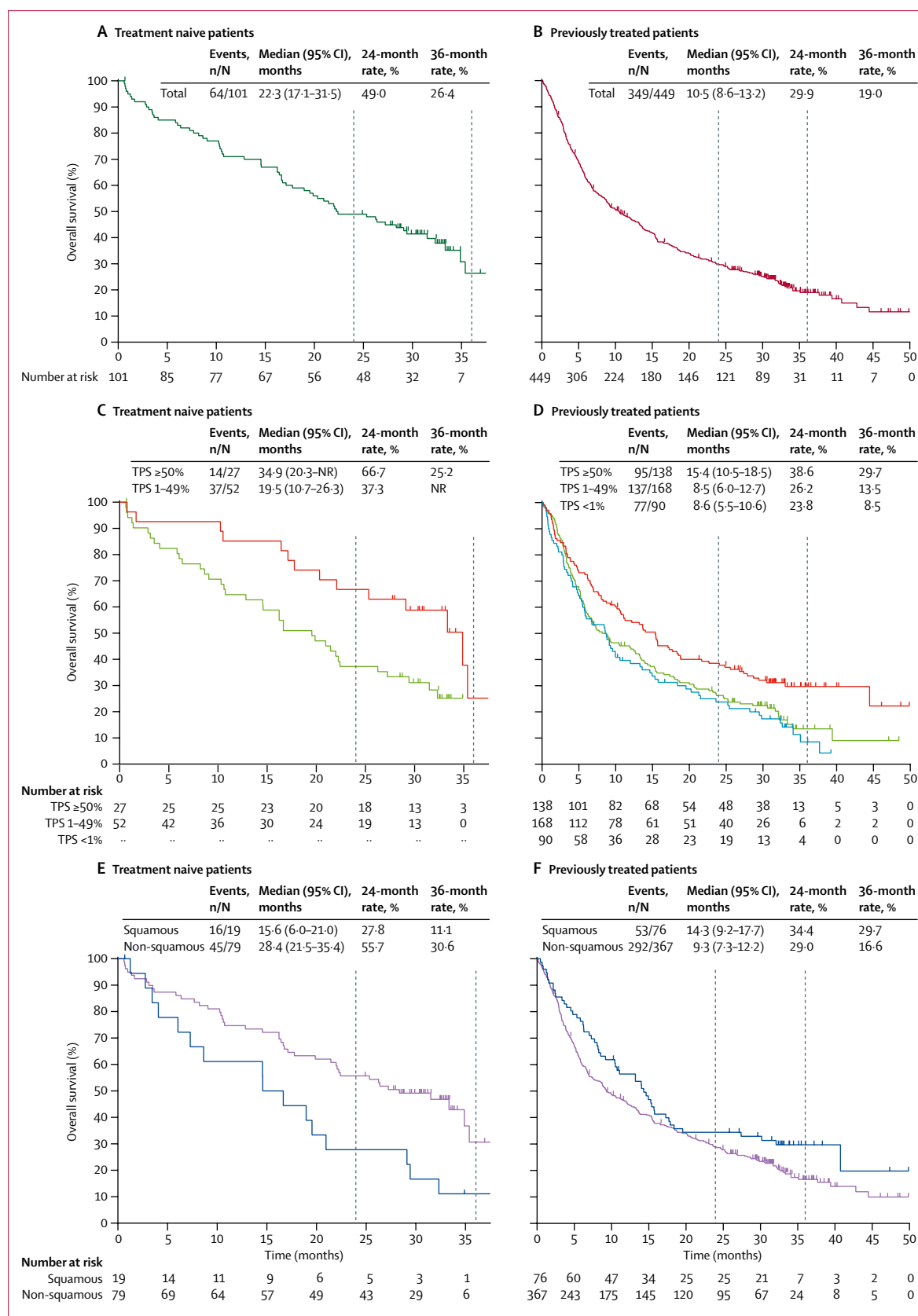


Figure 3: Kaplan-Meier estimates of overall survival
 (A) Treatment naive patients.
 (B) Previously treated patients.
 (C) Treatment naive patients by PD-L1 TPS ≥50% (red) and 1–49% (green).
 (D) Previously treated patients by PD-L1 TPS ≥50% (red), 1–49% (green), and <1% (blue).
 (E) Treatment naive patients by histology: squamous (blue); non-squamous (purple).
 (F) Previously treated patients by histology: squamous (blue); non-squamous (purple).
 n=number of patients who died. N=number of patients in the group or subgroup.
 NR=not reached.
 PD-L1=programmed death ligand 1. TPS=tumour proportion score.
 *PD-L1 TPS <1% group not presented owing to the small number of patients (n=12) initially deemed to have PD-L1 TPS >1% on the basis of the QualTek assay (Molecular Laboratories, Goleta, CA, USA), but later found to have PD-L1 TPS <1% by use of the Dako assay (Agilent Technologies, Carpinteria, CA, USA).

	Events, n/N	Median overall survival (months [95% CI])	36-month overall survival (% [95% CI])
Smoking history			
Ever smoked	245/324	12.2 (9.2–14.3)	21.9% (17.1–27.2)
Never smoked	104/125	7.6 (5.9–11.4)	11.9% (6.3–19.5)
Histology*			
Squamous	53/76	14.3 (9.2–17.7)	29.7% (19.6–40.4)
Non-squamous	292/367	9.3 (7.3–12.2)	16.6% (12.4–21.4)
EGFR status			
Mutant	63/74	6.0 (4.4–8.8)	6.7% (2.0–15.7)
Wild-type	272/355	12.0 (9.2–14.3)	21.1% (16.6–26.1)
Prior radiation			
Yes	189/236	9.1 (7.0–11.7)	18.4% (13.5–23.9)
No	160/213	13.2 (8.8–15.5)	19.5% (13.3–26.5)

n=number of patients who died. N=number of patients in the subgroup.
*Excluding patients with other or not specified histology (n=6).

Table 4: Overall survival in key subgroups of previously treated patients (n=449)

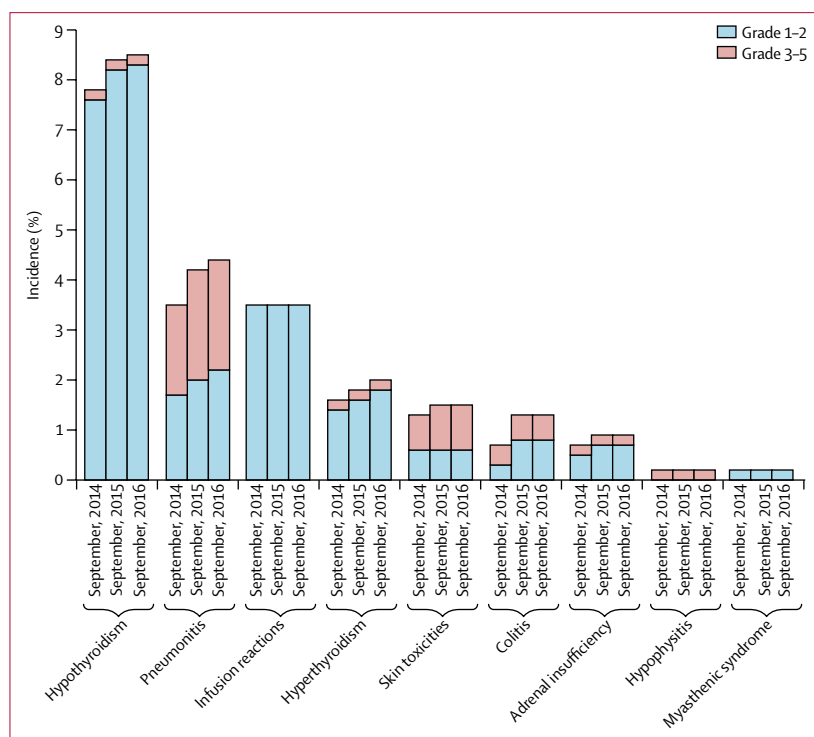


Figure 4: Incidence of immune-mediated adverse events and infusion reactions

Bar height indicates overall incidence of events (all grades) after 1 year (Sept 1, 2014),¹⁷ 2 years (Sept 18, 2015),¹⁶ and 3 years (Sept 1, 2016).

8.5 months (95% CI 6.0–12.7) for the 168 patients with tumour proportion score 1–49% and 8.6 months (5.5–10.6) for the 90 patients with tumour proportion score less than 1% (figure 3). Overall survival by select subgroups are provided in table 4 (previously treated) and the appendix (treatment naive). Median overall survival was 15.6 months (95% CI 6.0–21.0) for treatment naive

patients with squamous histology, 28.4 months (21.5–35.4) for treatment naive patients with non-squamous histology, 14.3 months (9.2–17.7) for previously treated patients with squamous histology, and 9.3 months (7.3–12.2) for previously treated patients with non-squamous histology (figure 3).

There were 83 (82%) progression events in the treatment naive group and 387 (86%) in the previously treated group. At data cutoff, all treatment naive patients had progressed or were censored prior to 36 months (Kaplan-Meier estimate of progression-free survival at 36 months not reached [95% CI NR]). The estimated 36-month progression-free survival was 9.2% (95% CI 6.1–13.0) for previously treated patients. Median progression-free survival was 10.3 months (95% CI 8.3–14.7) in the treatment naive cohort and 4.2 months (3.3–4.8) for previously treated patients. Kaplan-Meier curves for each group and by PD-L1 tumour proportion score are displayed in the appendix.

At the time of our analysis, 388 (71%) of 550 patients had had at least one treatment-related adverse event, including 66 (12%) of whom had at least one grade 3–5 treatment-related adverse event, and 30 (6%) who discontinued owing to a treatment-related adverse event. Similar results were observed in previously treated patients with *EGFR* mutations; there were too few treatment naive patients with *EGFR* mutations to evaluate. Overall, the most frequent treatment-related adverse events were fatigue (119 [22%] of 550; grade 1–2, 114 [21%] of 550; grade 3, 5 [1%] of 550), pruritus (63 [12%] of 550; all grade 1–2), and decreased appetite (59 [11%] of 550; grade 1–2, 58 [11%] of 550; grade 3, 1 [$<1\%$] of 550). The most frequent grade 3–4 treatment-related adverse events were pneumonitis (10 [2%] of 550) and fatigue (5 [1%] of 550). Serious adverse events were reported in 227 patients (41%), including 50 (9%) who had treatment-related serious adverse events. Two patients died because of treatment-related adverse events, including one with interstitial lung disease that occurred on day 12 and one with cardiopulmonary arrest on day 32. Immune-mediated adverse events and infusion reactions of any grade, regardless of attribution to therapy by investigators, occurred in 92 patients (17%), including 22 (4%) who had adverse events of grade 3–5 severity and 17 (3%) who discontinued treatment owing to immune-mediated adverse events and infusion reactions.

There was no evidence of late-onset grade 3–5 toxicity among the 550 patients included in the safety analyses. Between years 1–2 of study follow-up (September, 2014–September, 2015¹⁶), 14 additional patients developed grade 3–5 treatment-related adverse events; however, between years 2–3 (September, 2015–September, 2016), there were no additional treatment-related grade 3–5 adverse events; only one additional patient discontinued therapy because of a treatment-related adverse event. The incidence of immune-mediated adverse events and infusion reactions with pembrolizumab remained similar following year 1 (figure 4).

Discussion

This updated analysis of outcomes among patients enrolled in the phase 1 KEYNOTE-001 study represents the longest efficacy and safety follow-up reported so far with pembrolizumab for patients with advanced non-small-cell lung cancer. With median follow-up of approximately 3 years, the results suggest that pembrolizumab has long-term effects on objective response and survival for treatment naive and previously treated patients. The 36-month survival estimates were 26·4% for treatment naive and 19·0% for previously treated patients. In this analysis with longer follow-up (median 34·5 months) and fewer censored patients compared with the initial analysis of KEYNOTE-001 (median follow-up, 10·9 months), median overall survival was longer in both the treatment naive (22·3 vs 16·2 months) and previously treated (10·5 vs 9·3 months) groups compared with the previous report.⁶ Additionally, median progression-free survival was 10·3 months for treatment naive patients and 4·2 months for previously treated patients in our analysis. Additional evidence for the long-term benefits of anti-PD-1 therapy continues to accrue, with multiple studies showing improved survival with pembrolizumab treatment compared with chemotherapy, earlier in the course of disease. There was no evidence of cumulative immune-mediated toxicity with pembrolizumab treatment or of late-onset grade 3–5 toxicity.

The 36-month survival among patients with PD-L1 tumour proportion score of 50% or greater in treatment naive and previously treated groups is remarkable (>25%). Consistent with the initial report from KEYNOTE-001,⁶ median overall survival was increased among patients with tumour proportion score of 50% or greater versus patients with tumour proportion score 1–49% or less than 1%. Interestingly, although patients with higher PD-L1 expression are expected to have improved survival with pembrolizumab treatment,¹⁸ previously treated patients with tumour proportion score 1–49% seemed to have similar survival prevalence to those with tumour proportion score less than 1%. However, we acknowledge that in the context of a single-arm trial it is not possible to definitively evaluate the influence of differing levels of PD-L1 expression on survival and that this finding might be confounded by other prognostic factors, such as smoking history and *EGFR* or *ALK* status. Additionally, very few patients in the treatment naive group had tumour proportion score less than 1%, which limits interpretation (data not shown). Nevertheless, results from the randomised KEYNOTE-042 study in treatment naive patients and KEYNOTE-010 study in previously treated patients, similarly support the predictive value of PD-L1 tumour proportion score for survival, with the longest overall survival times among patients with tumour proportion score of 50% or greater and improved overall survival versus chemotherapy when including lower tumour proportion score groups ($\geq 20\%$ and $\geq 1\%$).^{9,11} Together, these results provide further support for the use of 22C3 antibody-based assays to evaluate PD-L1 expression when

selecting patients for treatment with pembrolizumab monotherapy, both in clinical trials and clinical practice. The PD-L1 IHC 22C3 pharmDx assay (Agilent Technologies, Carpinteria, CA, USA)¹⁹ companion diagnostic has been approved by the US Food and Drug Administration and is CE marked (ie, approved for marketing) in Europe for evaluation of PD-L1 expression in tumours.

Survival at 36 months was high across subgroups defined by smoking status and histology, although evaluation of outcomes in some subgroups should be interpreted with caution owing to the small number of patients and because this study was not designed or powered for these analyses. Notably, in the previously treated group, the association between smoking status and improved outcomes with pembrolizumab treatment reported in the initial analysis of KEYNOTE-001⁶ (higher proportions of patients with objective response among current or former smokers vs never smokers) persists in this updated analysis (longer survival among current or former smokers vs never smokers). However, our analysis did not show the same pattern among treatment naive patients, although this might have been influenced by the small number of patients who had never smoked in the treatment naive group ($n=11$). Additional research will be needed to establish the biological factors underlying the observed relationship between smoking status and treatment outcome.

Our analyses indicate improved overall survival in previously treated patients with wild-type *EGFR* status versus patients with *EGFR* mutations. After study initiation, the protocol was amended to require testing and exclusion of treatment naive patients with *EGFR* or *ALK* alterations; therefore, overall survival data are not available in these patients. Our findings are consistent with a meta-analysis that evaluated anti-PD-1 or anti-PD-L1 therapy in the second-line setting.²⁰ However, results from the phase 3 IMpower150 study²¹ provided evidence of improved progression-free survival with atezolizumab plus bevacizumab–carboplatin–paclitaxel versus bevacizumab–carboplatin–paclitaxel among patients with metastatic non-squamous non-small-cell lung cancer with *EGFR* or *ALK* alterations who had previously received treatment with an approved tyrosine kinase inhibitor. The role of anti-PD-1 or anti-PD-L1 therapy after first-line tyrosine kinase inhibitor treatment in patients with *EGFR* mutations is still an open question and is being evaluated in the KEYNOTE-789 study (ClinicalTrials.gov, NCT03515837).

Typically, responses occurred early during pembrolizumab therapy and were maintained over extended periods. Approximately half of the patients had an ongoing response at data cutoff. Because these data were assessed by investigators per immune-related response criteria (because central assessment had ceased), they cannot be directly compared with the proportion of patients with objective response by independent central

review per RECIST v1.1 previously reported for KEYNOTE-001.⁶ However, data before discontinuation of central assessment showed 92% concordance between the two methods. These data strongly suggest that patients who respond during pembrolizumab therapy have durable and clinically meaningful responses.

Late-onset toxicity was uncommon with pembrolizumab. The overall frequency of treatment-related toxicity and immune-mediated adverse events and infusion reactions was similar to that of previous reports,⁶ with low prevalence of treatment discontinuation. Together, these results provide the longest reported follow-up of pembrolizumab in advanced non-small-cell lung cancer so far, and support the safety and clinical tolerability of the drug in this setting. Data suggest that continuing treatment until toxicity or progressive disease might be a better approach than stopping treatment after 1 year.²² The optimal duration of therapy has not been well studied. The long-term safety data reported here, which include ten treatment naive and 38 previously treated patients who received treatment for at least 2 years, support the safety of long-term pembrolizumab treatment. Patient-reported outcomes were not evaluated in this study, but have been previously reported from KEYNOTE-024, which showed clinically meaningful improvement in health-related quality of life with pembrolizumab versus platinum-based chemotherapy as first-line treatment in patients with metastatic non-small-cell lung cancer and PD-L1 tumour proportion score of 50% or greater.²³

The decision to pool data for the pembrolizumab doses investigated in this study was based largely on the absence of difference in outcomes by dose previously observed in this study.⁷ This decision was further supported by studies that have compared various weight-based and fixed-dosing regimens, and shown similar exposure²⁴ and equivalence in efficacy and safety.²⁵ This includes data from KEYNOTE-010, which showed similar results between 2 and 10 mg/kg every 3 weeks in terms of overall survival, progression-free survival, clinical response, and safety and, more importantly, showed significantly improved efficacy and better tolerability with pembrolizumab (both doses) versus docetaxel.⁹

Our results provide important context for efficacy and safety assessments of pembrolizumab from the randomised, open-label phase 3 KEYNOTE-024,¹⁰ KEYNOTE-010,⁹ and KEYNOTE-042 studies.¹¹ These studies showed significantly improved overall survival with pembrolizumab monotherapy versus platinum-based chemotherapy or docetaxel after 11.2 months of follow-up for KEYNOTE-024, 13.1 months of follow-up for KEYNOTE-010, and 12.8 months of follow-up for KEYNOTE-042, with a significant proportion of patients censored at time of analysis (108 [35%] of 305 patients in KEYNOTE-024, 521 [50%] of 1033 in KEYNOTE-010, and 809 [64%] of 1274 in KEYNOTE-042 had died). Our results, which showed meaningful proportions of patients alive at a median follow-up of 3 years, suggest that patients treated

with pembrolizumab in these studies are likely to continue to derive benefit after longer follow-up. Two reports of updated analyses showed significantly longer overall survival with pembrolizumab versus chemotherapy in KEYNOTE-024²⁶ (median follow-up 25.2 months) and versus docetaxel in KEYNOTE-010⁵ (median follow-up 31 months).

Data from phase 1 and phase 3 studies of nivolumab and a phase 2 study of atezolizumab—all in patients with previously treated non-small-cell lung cancer—provide further support for the durability of benefit with anti-PD-1 or anti-PD-L1 therapy in patients with non-small-cell lung cancer. Overall survival with nivolumab in the phase 1 study was 18% after 3 years of follow-up and 16% after 5 years of follow-up.^{27,28} Similarly, in a pooled analysis after 3 years of follow-up in the phase 3 CheckMate 017 and CheckMate 057 studies, an overall survival benefit was shown with nivolumab versus docetaxel (HR, 0.70 [95% CI 0.61–0.81]). The 3-year overall survival for nivolumab versus docetaxel was 17% (14–21) versus 8% (6–11).²⁹ The 3-year survival for atezolizumab versus docetaxel in the phase 2 POPLAR study was 18.7% versus 10.0% ($p=0.042$).³⁰ To date long-term data are not available on outcomes with nivolumab or atezolizumab in treatment naive patients.

In summary, our results show that pembrolizumab provides long-term overall survival with a clinically tolerable safety profile for both treatment naive and previously treated patients with advanced non-small-cell lung cancer expressing PD-L1. These data illustrate the clinically meaningful benefits that immunotherapy provides in the treatment of non-small-cell lung cancer and suggest a potential for pembrolizumab to improve long-term survival among patients with advanced non-small-cell lung cancer.

Contributors

All authors contributed to study design; data collection, analysis, and interpretation; and the writing, review, and approval of the final manuscript. Funding for this research was provided by Merck Sharp & Dohme Corp, a subsidiary of Merck & Co, Inc, Kenilworth, NJ, USA, and their representatives contributed to various aspects of study design, data analysis and interpretation, and manuscript preparation. All authors had access to the data and had responsibility for the decision to submit this article for publication.

Declaration of interests

NBL has received honoraria for accredited CME lectures, unrelated to the submitted work, from AstraZeneca, Bristol-Myers Squibb, Pfizer, and Merck Sharp & Dohme. MDH has received research funding from Bristol-Myers Squibb; is a paid consultant to Merck, Bristol-Myers Squibb, AstraZeneca–MedImmune, Genentech–Roche, Janssen, Nektar, Syndax, Mirati, Shattuck Labs, Cowen, Inovio, Neon, and Novartis; received non-financial support from Bristol-Myers Squibb and AstraZeneca–MedImmune; and has a patent filed by MSK related to the use of tumour mutation burden to predict response to immunotherapy (PCT–US2015–062208), licensed to Personal Genome Diagnostics Inc. RH has received personal fees for participating in advisory boards for Merck Sharp & Dohme, AstraZeneca, Novartis, Roche, and Pfizer and speaker honorariums from Merck Sharp & Dohme, AstraZeneca, Novartis, Bristol-Myers Squibb, and Boehringer Ingelheim. EF served in a consulting and advisory role and participated in speaker's bureaus for AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Celgene, Eli Lilly, Guardant Health, Merck Sharp & Dohme, Novartis, Pfizer, Roche, and Takeda. LH has received grants from Merck, Xcovery, and

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Data sharing

Merck & Co, Inc's data sharing policy, including restrictions, is available at http://engagezone.merck.com/ds_documentation.php. Requests for access to the clinical study data can be submitted through the EngageZone site or via email to dataaccess@merck.com.

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